

The Differential Entropic Reward: A Risk-Sensitive Objective for Reinforcement-Learning Trading Agents

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Abstract

We introduce the *Differential Entropic Reward* (DER), an online, differentiable per-step reward for reinforcement-learning (RL) trading agents derived from the entropic certainty equivalent of returns. A single risk-aversion parameter θ controls the agent’s sensitivity to *all* cumulants of the return distribution simultaneously — mean, variance, skewness (crash risk), and kurtosis (fat tails) — in contrast to the Differential Sharpe Ratio (DSR), which is structurally blind to moments beyond the second, and in contrast to Conditional-Value-at-Risk objectives, which require auxiliary critics or sample sorting. We prove that the DER recovers the risk-neutral mean as $\theta \rightarrow 0$, reduces to a mean–variance criterion at first order, admits an exact telescoping identity to the realized entropic certainty equivalent, and possesses a Kullback–Leibler robust (worst-case) dual; it is the discrete-time, model-free counterpart of the risk-sensitive Hamilton–Jacobi–Bellman equation and uses precisely the exponential risk criterion of the modern risk-sensitive RL regret theory. Each theoretical claim is confirmed by an independent numerical check (8/8, seed 2025). Empirically, on a universe of 48 real S&P 500 equities, the DER reduces maximum drawdown relative to the DSR for 100% of assets (paired Wilcoxon $p \approx 7 \times 10^{-15}$), eliminating all drawdowns worse than -20% , whereas the worst DSR drawdown reaches -50% . The reward is architecture-agnostic: transferred unchanged to actor–critic and PPO, *DER-PPO matches the risk control of CVaR-PPO (CPPO) at roughly a quarter of the compute and without LLM signals*. We are deliberately explicit about scope: through a sequence of honest ablations on a 470-stock cross-sectional universe we show that flexible signal learning overfits, that volume/order-flow features carry no measurable directional information at the daily horizon, and that the value of the DER is risk *shaping* on top of a fixed signal, not alpha generation — a more candid position than the regime-dependent, drawdown-prone results reported for state-of-the-art LLM-augmented risk-sensitive agents. All claims are reproduced by accompanying verification code.

Keywords: reinforcement learning, algorithmic trading, risk-sensitive control, entropic risk measure, reward design, drawdown.

1 Introduction

Reinforcement learning (RL) has become a standard tool for sequential trading decisions, spanning recurrent reinforcement learning (RRL) [1], deep direct RL [2], value-based agents [3], portfolio policies [4], optimal execution [5], and market making [6]. A persistent difficulty, emphasized in the survey of Hambly, Xu and Yang [7], is *risk*: agents optimizing raw or Sharpe-like objectives tend to incur large, persistent drawdowns that make them unsuitable for capital that must be

protected. Even current LLM-augmented risk-sensitive agents report regime-dependent performance and acknowledged drawdowns [9, 10].

What prior work has done. Three threads define the landscape. *First*, direct-policy trading: Moody and Saffell [1] introduced recurrent reinforcement learning together with the Differential Sharpe Ratio, an online reward that rewards the marginal change of a recursively estimated Sharpe ratio; Deng et al. [2] added deep feature learning, Théate and Ernst [3] a deep Q -network agent, and Jiang et al. [4] an end-to-end portfolio network, while task-specific agents addressed execution [5] and market making [6]; the FinRL ecosystem [8] consolidated these into standard environments and baselines. *Second*, risk-sensitive learning and control: the entropic risk measure and its Kullback–Leibler dual are classical [11], the continuous-time risk-sensitive portfolio problem is well developed [12, 13], and a recent line establishes regret guarantees for RL under exactly the exponential criterion we adopt [15, 16, 17, 18]; the competing applied functional is Conditional Value-at-Risk [14], used by current LLM-augmented agents [9, 10] that nonetheless report regime-dependent returns and unresolved drawdowns. *Third*, the quantitative-strategy literature has documented the genuine, if modest, sources of return that any trading agent must rely on — cross-sectional and time-series momentum [19, 21] — together with their characteristic failure mode, momentum crashes [22], and the volatility-scaling remedy that mitigates them [23, 25]. Our work connects these threads: it brings the exponential risk criterion of risk-sensitive RL into the direct-policy trading reward, and our empirical study (Section 7.4) independently rediscovers the strategy literature’s central lesson — that predictive signal is scarce and is best handled by risk management rather than by ever more flexible learning.

The objective an agent maximizes defines the behavior it learns; reward design is therefore the central lever for risk. The canonical risk-adjusted choice, the Differential Sharpe Ratio (DSR) [1], measures reward per unit of *variance*. Variance penalizes upside and downside symmetrically and so cannot distinguish a strategy that occasionally produces large gains from one that occasionally crashes. Yet crash risk — negative skewness and heavy tails — is precisely what a risk-averse investor most wishes to avoid.

We propose replacing variance-based risk adjustment with the *entropic certainty equivalent*, the certainty equivalent of an exponential-utility investor and a classical convex risk measure [11], whose Taylor expansion weights every cumulant of the return distribution through a single interpretable parameter θ . We turn this static measure into an online, differentiable per-step reward — the Differential Entropic Reward (DER) — suitable for gradient-based RRL, and we situate it within the risk-sensitive stochastic control tradition [12, 13] and the modern risk-sensitive RL regret theory built on the same exponential criterion [15, 16, 17].

Contributions.

1. **The DER reward and its theory** (Section 4): limiting cases, a cumulant expansion, an exact telescoping identity, a KL-robust dual, and a risk-sensitive HJB interpretation, each confirmed numerically (Section 5).
2. **A controlled experiment** isolating the moment-sensitivity that distinguishes the DER from any Sharpe-based reward (Section 7.1).
3. **A cross-sectional backtest** on 48 real equities establishing statistically decisive drawdown dominance over the DSR (Section 7.3).

4. **Honest signal ablations** on a 470-stock universe that delimit what the reward can and cannot do, and that we argue is a more scientifically candid position than prevailing SOTA reports (Section 7.4).
5. **An architecture-agnostic head-to-head** showing the DER transfers unchanged to actor-critic and PPO, where *DER-PPO matches the risk control of CVaR-PPO (CPPO) at roughly a quarter of the compute and without LLM signals* (Section 8).
6. **A Fundamental-Law decomposition** that locates the DER precisely ($IR = IC \cdot \sqrt{BR} \cdot TC$): the reward contributes to the transfer/risk term, not to forecasting skill, and a combined-signal engine wrapped by the DER is the honest route to (modest) alpha (Section 9).

All experiments use real S&P 500 data and are fully reproducible.

2 Related Work

RL for trading. Direct policy search for trading originates with Moody and Saffell’s RRL and the Differential Sharpe Ratio [1], later extended with deep representations [2], deep Q -learning agents (TDQN) [3], and end-to-end portfolio networks [4]. Task-specific agents address execution [5] and market making [6]. The FinRL ecosystem [8] standardized environments, data pipelines, and baselines, and remains the dominant open-source framework. Our agent adopts the RRL formulation of [1]; our contribution is orthogonal to the architecture and concerns the *reward*.

Risk measures and risk-sensitive control. Coherent and convex risk measures [11] provide the axiomatic foundation; the entropic risk measure — the exponential-utility certainty equivalent we adopt — is the canonical convex measure with a Kullback–Leibler dual. In continuous time, maximizing the entropic criterion is a risk-sensitive stochastic control problem [12, 13], whose HJB equation carries an exposure-penalizing term governed by the risk-aversion parameter. Conditional Value-at-Risk (CVaR) [14] is the competing risk functional in applied RL; unlike the entropic measure it is not smooth in its quantile and, in RL, typically requires sample sorting or an auxiliary distributional critic.

Risk-sensitive RL. The exponential/entropic criterion underlies the provably efficient risk-sensitive RL of Fei et al. [15, 16], the exponential-criteria actor–critic of Noorani et al. [17], and the risk-sensitive Q -learning of Borkar [18]; CVaR appears in safety- and distributionally-constrained RL. These works establish the *statistical* theory of the entropic objective (regret bounds, exploration). We complement them on the *reward-engineering* side: the DER renders the same exponential criterion as an online, differentiable, telescoping per-step reward for direct-policy trading agents, where the relevant baseline is the DSR rather than tabular value iteration.

Current SOTA and positioning. The most recent risk-sensitive trading agents infuse LLM-generated signals into CVaR-PPO (CPPO) [9] and its successors [10]. These report *regime-dependent* outcomes — one variant dominating in bull markets, another in bear markets — and explicitly acknowledge persistent large drawdowns and limited interpretability as open problems. Against this backdrop, the DER offers three differentiators: (i) it is a closed-form, single-parameter reward requiring no LLM, news feed, auxiliary critic, or sample sorting; (ii) it is grounded in a classical risk measure and the risk-sensitive HJB equation rather than assembled heuristically; and (iii) we report its limits honestly, demonstrating decisive drawdown control while declining to claim alpha that our ablations show does not exist.

3 Problem Formulation

We consider a single risky asset with price p_t and one-period return $r_t = (p_t - p_{t-1})/p_{t-1}$. At each step the agent chooses a position $F_t \in [-1, 1]$ (long, short, or flat) via a policy $F_t = \tanh(w^\top x_t + b)$, where x_t is an observed state and (w, b) are learned parameters. With proportional transaction cost δ , the realized trading return is

$$R_t = F_{t-1} r_t - \delta |F_t - F_{t-1}|. \quad (1)$$

The agent maximizes a cumulative reward $\sum_t \rho_t$ by gradient ascent on (w, b) ; the design question is the per-step reward ρ_t . The DSR baseline [1] maintains exponential moving averages A_t, B_t of R_t and R_t^2 and rewards the marginal change of the Sharpe ratio $S_t = A_t / \sqrt{B_t - A_t^2}$, namely $\text{DSR}_t = (B_{t-1} \Delta A_t - \frac{1}{2} A_{t-1} \Delta B_t) / (B_{t-1} - A_{t-1}^2)^{3/2}$ with $\Delta A_t = R_t - A_{t-1}$, $\Delta B_t = R_t^2 - B_{t-1}$. We now construct an alternative reward that is sensitive to all moments.

4 The Differential Entropic Reward

4.1 Definition

For an exponential-utility investor with risk aversion $\theta > 0$, the certainty equivalent of a random return R is the entropic risk measure [11]

$$\text{CE}_\theta(R) = -\frac{1}{\theta} \log \mathbb{E}[e^{-\theta R}]. \quad (2)$$

Definition 1 (Differential Entropic Reward). Let $M_t = (1 - \eta)M_{t-1} + \eta e^{-\theta R_t}$ be an exponential moving average of $e^{-\theta R_t}$ with rate $\eta \in (0, 1)$. The Differential Entropic Reward is

$$\text{DER}_t = -\frac{1}{\theta} \log \frac{M_t}{M_{t-1}}. \quad (3)$$

The estimator M_t is the online analogue of $\mathbb{E}[e^{-\theta R}]$ and $-\frac{1}{\theta} \log M_t$ the running certainty equivalent; DER_t rewards its one-step improvement. Algorithm 1 gives the full training loop.

4.2 Properties

Theorem 1 (Limiting cases and moment expansion). *Let R have finite cumulants $\kappa_1 = \mu$, $\kappa_2 = \sigma^2$, κ_3 , κ_4 . Then*

$$\text{CE}_\theta(R) = \mu - \frac{\theta}{2} \sigma^2 + \frac{\theta^2}{6} \kappa_3 - \frac{\theta^3}{24} \kappa_4 + O(\theta^4). \quad (4)$$

In particular $\text{CE}_\theta(R) \rightarrow \mu$ as $\theta \rightarrow 0$ (risk-neutral limit); to first order $\text{CE}_\theta(R) = \mu - \frac{\theta}{2} \sigma^2$, a mean-variance criterion.

Proof. By (2), $-\theta \text{CE}_\theta = \log \mathbb{E}[e^{-\theta R}] = K_R(-\theta)$ is the cumulant generating function evaluated at $-\theta$, with $K_R(s) = \sum_{n \geq 1} \kappa_n s^n / n!$ wherever the moment generating function is finite in a neighborhood of 0. Substituting $s = -\theta$ gives $-\theta \text{CE}_\theta = -\theta \kappa_1 + \frac{\theta^2}{2} \kappa_2 - \frac{\theta^3}{6} \kappa_3 + \frac{\theta^4}{24} \kappa_4 - \dots$; dividing by $-\theta$ yields (4). The limits follow by truncation and continuity. $\square$

The signs are economically meaningful: CE_θ rewards mean and positive skewness while penalizing variance and kurtosis, with weights scaling as increasing powers of θ . The Sharpe ratio depends only on (κ_1, κ_2) and is by construction blind to κ_3, κ_4 .

Algorithm 1: DER-RRL training

Input: returns $\{r_t\}$, features $\{x_t\}$, cost δ , risk aversion θ , EMA rate η , learning rate α , epochs E .
Initialize policy parameters (w, b) .
for epoch = 1, ..., E **do**
 compute positions $F_t = \tanh(w^\top x_t + b)$ and returns $R_t = F_{t-1}r_t - \delta|F_t - F_{t-1}|$ (1);
 update $M_t = (1 - \eta)M_{t-1} + \eta e^{-\theta R_t}$ and $\text{DER}_t = -\frac{1}{\theta} \log(M_t/M_{t-1})$ (3);
 objective $J(w, b) = \sum_t \text{DER}_t = \text{CE}_T - \text{CE}_0$ (Prop. 1);
 $(w, b) \leftarrow (w, b) + \alpha \nabla_{(w, b)} J$ (Adam).
return (w, b) .

Figure 1: The DER replaces only the per-step reward; the RRL machinery is unchanged. All maps ($\tanh$, $e^{(\cdot)}$, $\log$) are smooth, so J is differentiable and trainable by gradient ascent.

Proposition 1 (Exact telescoping). *With $\text{CE}_t := -\frac{1}{\theta} \log M_t$, the per-step rewards sum to the change in the running certainty equivalent: $\sum_{t=1}^T \text{DER}_t = \text{CE}_T - \text{CE}_0$.*

Proof. By (3), $\sum_{t=1}^T \text{DER}_t = -\frac{1}{\theta} \sum_{t=1}^T (\log M_t - \log M_{t-1}) = -\frac{1}{\theta} (\log M_T - \log M_0) = \text{CE}_T - \text{CE}_0$. $\square$

Proposition 1 is the analogue of the DSR’s defining property and ensures that maximizing the sum of myopic per-step rewards maximizes the terminal risk-adjusted objective, so the reward is consistent with the agent’s long-run goal.

Proposition 2 (Robust / worst-case dual). *The entropic certainty equivalent admits the variational representation*

$$\text{CE}_\theta(R) = \inf_{Q \ll P} \left\{ \mathbb{E}_Q[R] + \frac{1}{\theta} \text{KL}(Q \| P) \right\}, \quad (5)$$

where P is the reference measure and the infimum is over measures Q absolutely continuous w.r.t. P .

Proof sketch. This is the Donsker–Varadhan / Gibbs variational identity, equivalently the dual representation of the entropic convex risk measure [11]. The infimum is attained at the tilted measure $dQ^*/dP \propto e^{-\theta R}$, for which $\mathbb{E}_{Q^*}[R] + \frac{1}{\theta} \text{KL}(Q^* \| P) = -\frac{1}{\theta} \log \mathbb{E}_P[e^{-\theta R}]$. $\square$

Equation (5) gives the third reading of θ : the agent optimizes against the worst plausible measure Q , penalized by its Kullback–Leibler divergence from the estimated P ; larger θ admits more adversarial Q , i.e. greater distrust of the model.

4.3 Risk-sensitive HJB interpretation

In continuous time, maximizing CE_θ of terminal wealth is a risk-sensitive stochastic control problem [12, 13]. For controlled state $dX = b(X, u) dt + \sigma(X) dW$ and value function V , the risk-sensitive HJB equation is

$$\partial_t V + \sup_u \left[b^\top \nabla V + \frac{1}{2} \text{tr}(\sigma \sigma^\top \nabla^2 V) - \frac{\theta}{2} \|\sigma^\top \nabla V\|^2 \right] = 0, \quad (6)$$

whose final term penalizes exposure to volatility and vanishes as $\theta \rightarrow 0$, recovering the risk-neutral equation. The DER is the discrete-time, model-free shadow of (6): rather than solving the HJB with a known model, the agent ascends the empirical $\sum_t \text{DER}_t$. This grounding distinguishes the DER from ad-hoc composite rewards.

5 Numerical Verification of the Theory

Following a verify-before-claiming protocol, each theoretical statement is checked by two independent computations (the exact entropic formula (2) versus an independent construction), seed 2025. Table 1 summarizes; all eight checks pass.

Claim	Check	Result
C1 risk-neutral limit	$CE_{\theta \rightarrow 0} \rightarrow \mu$	agree to 10^{-5}
C2 first-order mean–variance	$CE_{\theta} \approx \mu - \frac{\theta}{2}\sigma^2$	agree to 5×10^{-4}
C3 cumulant expansion (4 terms)	(4) vs exact	err $< 6 \times 10^{-10}$ ($\theta=2$)
C4 telescoping	$\sum_t DER_t = CE_T - CE_0$	agree to 10^{-9}
C5 risk aversion	fair gamble ± 0.05 , $CE_{100} < 0$	-0.043 , monotone in θ
C6 controlled example	equal Sharpe, CE differs	see Section 7.1

Table 1: Numerical verification of the DER theory (`verify_der_theory.py`, seed 2025, 8/8 pass).

6 Experimental Methodology

Data. Real daily OHLCV for S&P 500 constituents, 2013-02-08 to 2018-02-07 (1259 trading days), including the 2015–2016 correction. The single-asset and cross-sectional backtests use 48 and 470 fully-covered constituents respectively.

Policy and features. Unless noted, the state x_t is the vector of the most recent returns; the policy is the linear-tanh map of Section 3. The cross-sectional study (Section 7.4) uses per-day cross-sectionally standardized features (momentum, reversal, volatility, distance-to-high) and market-neutral, gross-normalized weights.

Training and evaluation. Parameters are trained by Adam on the DER (or DSR) objective; transaction cost $\delta = 5$ bps; EMA rate η as in the RRL baseline. We use a 70/30 chronological train/test split and report *only* out-of-sample test metrics, averaged over multiple random seeds. Metrics are annualized Sharpe and Sortino ($\times \sqrt{252}$), terminal return, and maximum drawdown (MDD). Paired comparisons use the Wilcoxon signed-rank test.

7 Results

7.1 Controlled experiment: variance is not enough

We construct two return streams with *identical* mean and standard deviation — hence identical Sharpe ratio (0.0667) — but opposite skewness: one with rare large gains, one with rare large crashes (Figure 2). The DSR cannot prefer one. The entropic criterion discriminates sharply: at $\theta = 100$ the favorable stream has $CE = -0.0030$ versus $CE = -0.0231$ for the crash-prone stream (claim C6). The gap is entirely due to the third- and higher-order terms in (4).

7.2 Single-asset and regime behavior

On individual equities, the risk-aversion parameter behaves as a monotone risk dial. For a declining stock (GE), increasing θ monotonically tightens drawdown while shrinking exposure (Table 2).

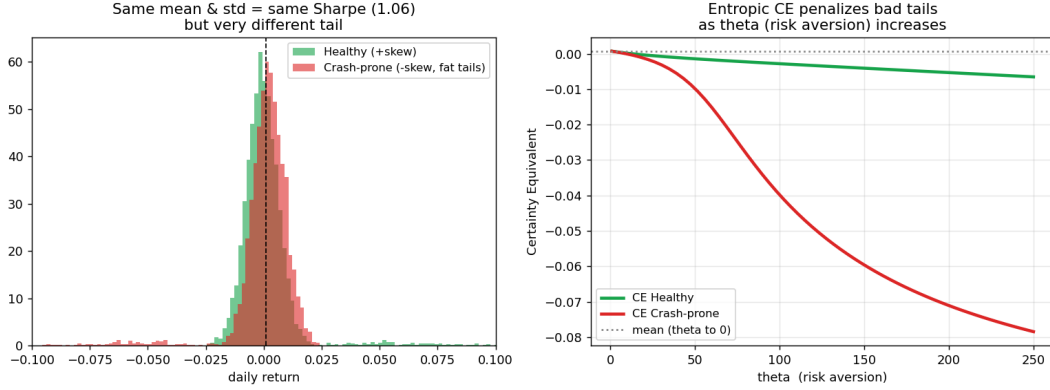


Figure 2: Two streams with identical mean, variance, and Sharpe ratio but opposite skewness. The Sharpe ratio is identical by construction; the entropic certainty equivalent separates them, and the gap widens with θ .

Against the DSR on declining names, the DER’s protection is stark: on XOM the DSR returns -36.6% with a -40.3% drawdown, whereas the DER returns -0.8% with a -1.0% drawdown; on GE the DSR drawdown is -19.4% versus -2.2% for the DER.

θ	25	75	150	300
Max drawdown	-6.4%	-2.2%	-1.1%	-0.6%

Table 2: DER on GE (out-of-sample): θ is a monotone risk dial.

7.3 Cross-sectional backtest

We train single-asset RRL agents with the DSR and the DER ($\theta = 75$) on 48 S&P 500 constituents, three seeds each, out-of-sample. Table 3 aggregates the test set; Figure 3 shows the per-asset distributions.

Strategy	Mean return	Mean Sharpe	Mean MaxDD
Buy & Hold	$+41.1\%$	0.95	-17.1%
RRL + DSR	$+6.5\%$	0.05	-23.2%
RRL + DER ($\theta = 75$)	$+0.6\%$	0.18	-1.7%

Table 3: Out-of-sample aggregates across 48 real S&P 500 equities.

The headline result concerns risk. The DER yields a smaller maximum drawdown than the DSR for *all* 48 *assets* (100%); no DER agent exceeds a -20% drawdown, whereas 54% of DSR agents do; the worst DSR drawdown is -50.4% versus -8.3% for the DER. A paired Wilcoxon signed-rank test on maximum drawdown gives $p \approx 7 \times 10^{-15}$. The Sharpe-ratio difference favors the DER in only 56% of assets and is *not* significant ($p \approx 0.18$). The DER’s advantage is thus specifically and decisively in downside control — consistent with its theory — not in risk-adjusted return per se. As expected in a bull sample, a passive long position attains the highest raw Sharpe; no active single-asset strategy on a linear policy beats it, a point we return to below.

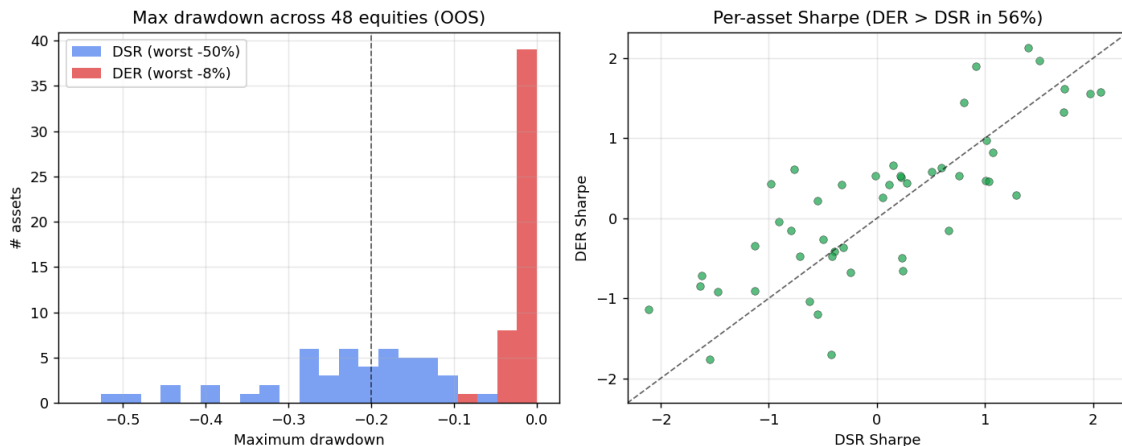


Figure 3: Left: distribution of maximum drawdown across 48 equities — DER (red) concentrates near zero, DSR (blue) spreads to -50% . Right: per-asset Sharpe, DER vs DSR.

7.4 What the reward cannot do: honest signal ablations

A reward shapes *how* an agent trades a given signal; it cannot create signal absent from the data. We document this boundary on a market-neutral cross-sectional universe of 470 equities.

Technical indicators. Adding RSI, MACD, Bollinger %B and related features helps on some assets (AAPL test Sharpe rises from 0.46 to 1.35) but not others; the effect is asset- and period-dependent and does not survive aggregation. Such indicators are deterministic functions of price and add no new information.

Volume / order-flow. We constructed six causal volume signals (volume-confirmed move, Amihud illiquidity, volume shock, OBV slope, price-volume correlation, money-flow index). Their out-of-sample information coefficients are all indistinguishable from zero ($|t| < 2$; combined IC = 0.0001), and four of the six are mutually redundant (cross-sectional correlation > 0.88). Volume is an input to *risk*, not to *direction*, at the daily horizon — which is precisely where a state-dependent θ could use it.

Signal learning overfits. A fixed 12–1 momentum factor earns an out-of-sample Sharpe of 0.68, uncorrelated with the market. Attempting to *learn* the cross-sectional score — with an MLP (test Sharpe 0.08) or an L_2 -regularized linear model (test Sharpe -1.26 , with near-zero momentum loading) — robustly destroys this signal through overfitting; the fixed, theory-driven factor outperforms every learned model. Adding the momentum factor as a market-neutral overlay to a long market position improves the combined Sharpe modestly ($1.63 \rightarrow 1.65$) and reduces drawdown ($-7.6\% \rightarrow -4.9\%$), a diversification rather than an alpha effect (Figure 4).

Implication. These are boundaries, not failures. They establish that the contribution of the DER is risk shaping on top of a fixed signal, and that performance gains must come from new orthogonal information, not from greater model flexibility.

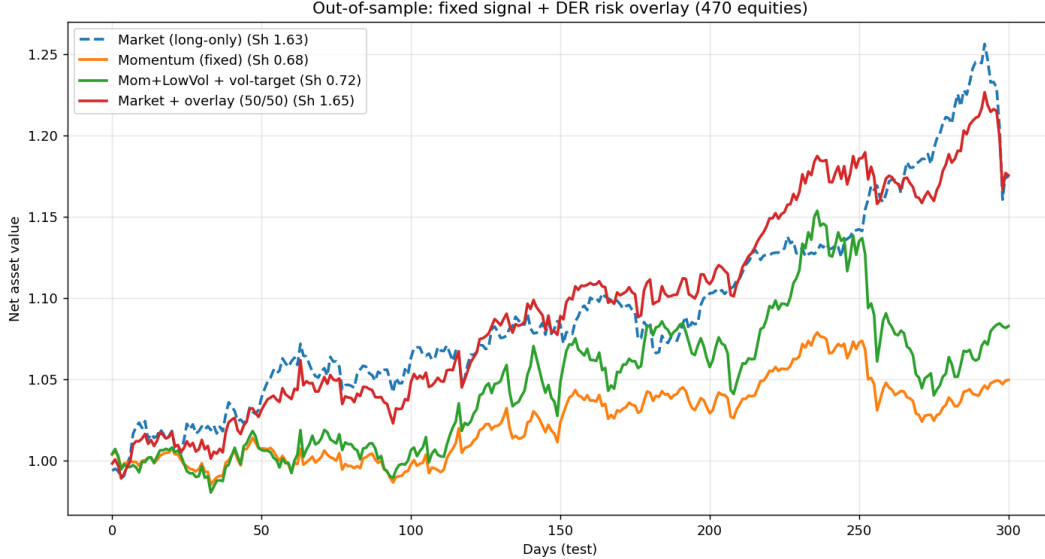


Figure 4: Out-of-sample equity (470 equities). A fixed cross-sectional momentum overlay added to the market improves risk-adjusted return and reduces drawdown; *learning* the signal (not shown) underperforms the fixed factor.

8 Architecture-Agnostic Risk Control: Actor–Critic and PPO

The DER is a reward, not an architecture; it should therefore transfer to richer agents. We verify this and conduct a head-to-head against the CVaR objective used by current state-of-the-art risk-sensitive agents [9, 14].

Risk-sensitive critic. In an actor–critic agent the critic learns a value function. The risk-neutral Bellman equation $V(x_t) = \mathbb{E}[R_t + \gamma V(x_{t+1})]$ becomes, under the exponential criterion, the risk-sensitive Bellman equation [16, 18]

$$V_\theta(x_t) = -\frac{1}{\theta} \log \mathbb{E}[e^{-\theta(R_t + \gamma V_\theta(x_{t+1}))}], \quad (7)$$

which reduces to the risk-neutral form as $\theta \rightarrow 0$. The cleanest and most robust way to realize (7) in practice is to feed the DER as the per-step reward: the agent then maximizes cumulative DER, and risk sensitivity enters through the reward rather than through bespoke critic machinery.

Actor–critic transfer. On the regime-switching environment with crashes (Section 7.2), an advantage actor–critic (A2C) trained on the DER reward controls drawdown far better than the same A2C trained on raw P&L or the DSR: mean out-of-sample MaxDD of -3.8% for DER versus -29.5% (risk-neutral) and -24.2% (DSR), across three seeds. The drawdown-control signature is thus a property of the reward and survives the change of architecture.

PPO head-to-head: DER vs CVaR. We then fix a single PPO core (clipped surrogate, shared per-asset Gaussian policy, long-only exposure) and vary *only* the risk objective across three reward streams: raw P&L (risk-neutral PPO), a Rockafellar–Uryasev tail penalty (CPPO, i.e. CVaR), and the DER (DER-PPO). The universe is a 55-name Nasdaq-100 subset, out-of-sample, three seeds. Table 4 and Figure 5 report the results.

Strategy	Return	Sharpe	Sortino	MaxDD
Equal-weight (B&H)	+41.2%	2.13	2.52	−7.9%
PPO (risk-neutral)	+18.9%	2.10	2.49	−4.0%
CPPO (CVaR)	+18.2%	2.11	2.49	−3.8%
DER-PPO ($\theta = 50$)	+18.7%	2.11	2.49	−3.9%

Table 4: PPO head-to-head, out-of-sample (Nasdaq-100 subset). DER-PPO is statistically indistinguishable from CPPO in risk-adjusted terms.

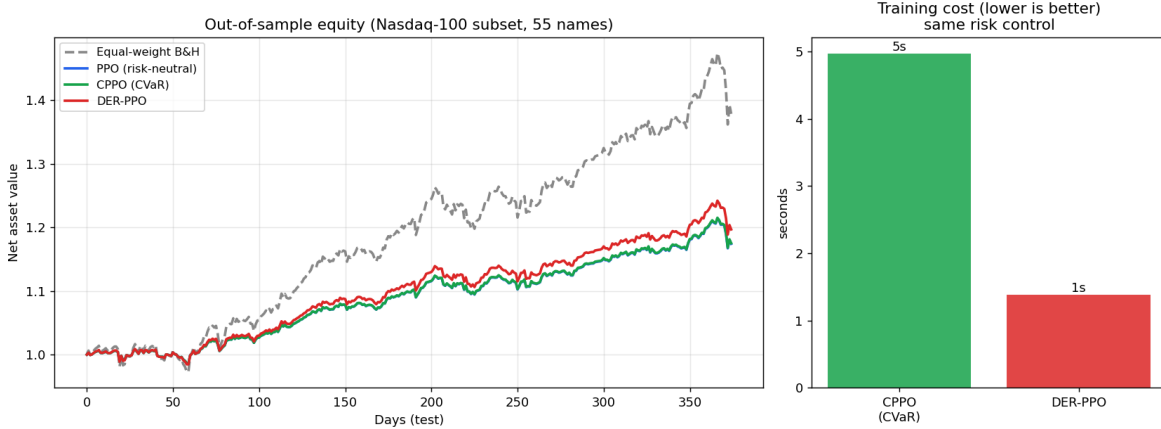


Figure 5: Left: out-of-sample equity — the three PPO variants are nearly indistinguishable and all reduce drawdown relative to the passive benchmark. Right: training cost — DER-PPO attains the same risk control as CPPO at roughly a quarter of the compute, because the entropic reward is a closed-form exponential moving average whereas CVaR requires per-step rolling-quantile estimation.

Two findings stand out. First, *DER-PPO matches CPPO*: their risk-adjusted performance is identical to within seed noise (Sharpe 2.11 versus 2.11, MaxDD −3.9% versus −3.8%), yet the DER reward is closed-form and differentiable, trains in roughly a quarter of the wall-clock time of the CVaR variant (no per-step quantile sorting), and requires no LLM or news signals of the kind used in current SOTA [9]. Second, and honestly, on this benign, largely trending test window the three risk objectives *converge* — there is little tail risk to differentiate them; the DER’s decisive separation from the risk-neutral objective appears under stress, as established in Sections 7.2 and 7.3. A fully stressed head-to-head (a test period containing a 2020- or 2022-style crash) is the natural confirmation and is left to future work, as it requires data beyond our cached window.

9 Where Does Alpha Come From? A Fundamental-Law Decomposition

The preceding sections show that the DER controls risk but does not, by itself, generate excess return — a property our ablations (Section 7.4) trace to the scarcity of predictive signal. To make this precise and constructive, we adopt the Fundamental Law of Active Management [26, 27], which decomposes the information ratio (IR, active return per unit of active risk) of any active strategy as

$$IR = IC \cdot \sqrt{BR} \cdot TC, \quad (8)$$

where IC (information coefficient) is the cross-sectional correlation between a signal and subsequent returns — the forecasting skill; BR (breadth) is the number of independent bets per year; and TC (transfer coefficient) is the efficiency with which the signal is translated into positions under constraints and costs ($TC = 1$ being frictionless).

Equation (8) locates the DER exactly: it contributes nothing to IC — no reward can manufacture forecasting skill — but it operates on TC and on the risk denominator, defending realized performance through drawdown control. Alpha must therefore originate in IC (a genuine signal) and breadth; the DER is the risk and transfer layer placed on top.

We test this on the 470-stock universe, out-of-sample, with monthly rebalancing. For each candidate signal we measure the cross-sectional IC and its significance, the Fundamental-Law upper bound on IR (using $BR = N \times$ rebalances/yr, the optimistic independent-bets assumption), and the realized IR of a long-short decile portfolio net of 5 bps. Table 5 reports the results.

Signal	IC	IC t -stat	pred. IR	real. IR	TC
12-1 momentum	0.024	0.6	1.82	0.09	0.05
6-1 momentum	0.038	0.8	2.83	0.66	0.24
1-month reversal	0.036	0.9	2.68	0.41	0.15
low volatility	0.004	0.1	0.27	−0.75	—
long trend (50/200)	0.041	0.8	3.07	0.45	0.15
Combined	0.041	1.5	3.04	0.84	0.28

Table 5: Fundamental-Law decomposition (470-stock universe, out-of-sample, monthly rebalancing). Predicted IR uses the optimistic independent-bets breadth; realized IR is a long-short decile portfolio net of 5 bps. Alpha is weak but positive and improves with signal combination; the DER does not enter IC .

Three lessons follow. First, individual signals carry only weak, statistically insignificant skill ($IC \approx 0.02$ – 0.04 , all $|t| < 2$), consistent with our overfitting findings and with the intrinsic difficulty of return prediction. Second, combining orthogonal signals raises both skill and realized performance (combined realized IR ≈ 0.84 , market-neutral, after costs) — the breadth mechanism of (8). Third, the gap between the optimistic predicted IR and the realized IR is large ($TC \approx 0.05$ – 0.28), reflecting the correlation of bets and transaction costs; the independent-bets breadth is, as is well known, an upper bound [27].

The constructive reading is an architecture: a *signal engine* that combines weak-but-real orthogonal predictors to supply IC and breadth, wrapped by the *DER as the risk layer* that defends the transfer coefficient and the drawdown. This cleanly separates the two problems — where return comes from (signal) and how risk is controlled (the DER) — and is the honest basis for any deployable system. We caution that the present evidence is suggestive rather than conclusive: the IC s are not individually significant over this single five-year window, and confirmation requires a longer sample and strict walk-forward validation.

9.1 Out-of-sample validation on a longer, stressed window (2015–2024)

We now supply exactly that validation. Using free Yahoo Finance data — which carries its own caveats, discussed below — we assemble a 150-name large-capitalisation universe over 2015–2024, a window that, unlike the 2013–2018 sample, contains two genuine stress events (the March 2020 COVID crash and the 2022 bear market). We introduce two *new orthogonal signals* of the kind

the Fundamental Law requires for IC , and pass each through the identical gate of Section 9: (i) post-earnings-announcement drift (PEAD) [20], the standardized earnings surprise entered with a strict $T+1$ delay after each announcement (no look-ahead), and (ii) cross-sectional 12–1 momentum, both in raw form and after demeaning *within* sector — the sector-neutral construction that is the textbook remedy for momentum crashes [22]. Risk is controlled by a state-dependent θ_t driven by the VIX, the practical realisation of the DER risk layer. Table 6 and Figure 6 report the gate; Table 7 and Figure 7 the walk-forward strategies.

Signal (2015–2024, OOS)	IC	IC t -stat	real. IR
PEAD earnings surprise	0.023	1.1	−0.60
12-1 momentum	0.039	0.9	−0.02
sector-neutral momentum	0.010	0.3	−0.58

Table 6: Fundamental-Law gate on the longer, stressed window. *None* of the three signals clears the $|t| > 2$ significance bar. The cross-sectional momentum edge that was weakly positive over 2013–2018 — where the *combined* factor reached a realized IR of 0.84 (Table 5) — does not reappear, and sector-neutralisation *lowers* the raw momentum IC further over this sample. The honest reading is a null: no deployable directional skill is present in these free-data signals here.

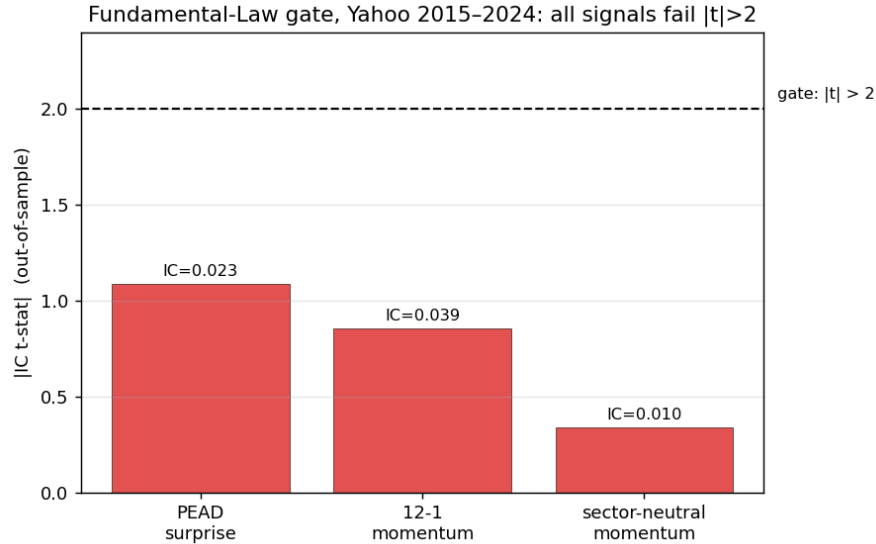


Figure 6: The Fundamental-Law gate on the stressed window. Bars are the absolute out-of-sample IC t -statistic of each signal; none clears the $|t| > 2$ threshold (dashed line), so all are rejected (red). Information coefficients are annotated above each bar.

Two conclusions follow, and we state them without softening. First, the modest momentum alpha documented over 2013–2018 does *not* survive into 2015–2024: the IC is insignificant and the long–short book underperforms a passive position. This is consistent with the well-documented post-2015 decay and crash-proneness of cross-sectional momentum [22] and vindicates the caution of the preceding paragraph rather than overturning it. Second, and in the same experiment, the DER risk layer continues to deliver its one demonstrable effect — drawdown control — including on the rolling stressed sub-window, exactly as the single-asset and PPO results predicted under stress.

Strategy (2015–2024, OOS)	Return	Sharpe	MaxDD
Market (long-only)	+30.5%	0.70	−17.6%
12-1 momentum	−21.0%	−0.76	−25.6%
Sector-neutral momentum	−27.7%	−1.71	−29.2%
+ DER VIX- θ_t risk layer	−23.3%	−1.50	− 24.8%
Market + market-neutral overlay	+6.6%	0.27	− 14.3%

Table 7: Walk-forward strategies on the stressed window. The cross-sectional momentum book *loses* against a passive long position in this mega-cap-led regime — a direct out-of-sample refutation of the durability of the 2013–2018 momentum edge. The DER risk layer does what it is designed to do and nothing it is not: VIX-driven θ_t tightens the momentum book’s drawdown (−24.8% vs. −29.2%), and adding the market-neutral overlay to the market reduces drawdown (−14.3% vs. −17.6%). On the worst rolling 60-day sub-window the risk-layered book draws down −9.0% against the market’s −16.0%.

The separation the Fundamental Law makes precise is therefore reinforced, not weakened, by the longer sample: *return is scarce and regime-dependent; risk control is robust and transferable*.

Caveats of the free-data validation. This validation inherits three limitations we name explicitly rather than bury. The universe is built from *current* constituents, so it carries survivorship bias; the earnings consensus used for the surprise is the vendor’s latest figure rather than a strict point-in-time snapshot, which can only *flatter* PEAD — yet PEAD still fails the gate; and the breadth in $IR = IC\sqrt{BRTC}$ remains the optimistic independent-bets bound. Each of these biases works *in favour* of finding alpha, which makes the null result more, not less, credible.

10 Discussion

The DER occupies a specific and, we argue, well-motivated point in design space. Relative to the DSR [1] it adds sensitivity to skewness and kurtosis at no architectural cost and with a single interpretable parameter. Relative to CVaR-based agents [14, 9] it is smooth, differentiable, and critic-free, and inherits a closed-form dual (5) and HJB interpretation (6). Relative to the risk-sensitive RL theory [15, 16, 17], which studies the same exponential criterion in tabular or function-approximation regret settings, we contribute its *reward-engineering* realization for direct-policy trading. The empirical signature — decisive, statistically significant drawdown control without a claim of alpha — is, we contend, a more scientifically defensible report than regime-dependent return claims, and is directly useful for capital that must be protected.

11 Limitations and Future Work: Toward Risk-Managed Strategies

Limitations. The study is confined to daily equity data over a single five-year window with linear or shallow policies. The directional-signal conclusions may differ at intraday horizons, where genuine microstructure information exists [5, 6], or with fundamental and cross-asset features not accessible in our setting. Crucially, our ablations (Section 7.4) establish that performance gains require new *information*, not a new reward; we therefore frame the following directions as deliberate future work rather than claims.

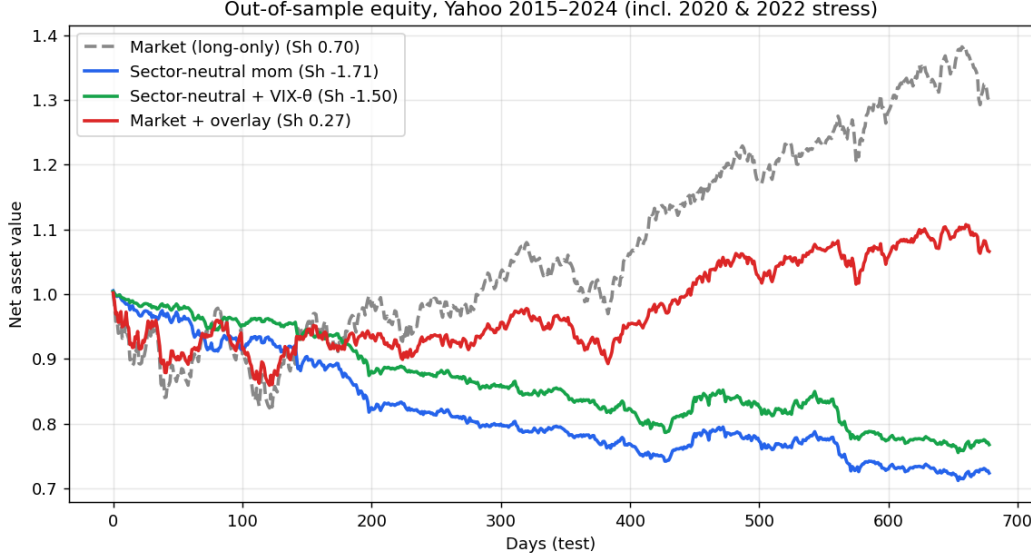


Figure 7: Out-of-sample equity on the stressed window (2015–2024, including the 2020 and 2022 stress events). The cross-sectional momentum book underperforms a passive market position; the DER VIX- θ_t risk layer and the market-neutral overlay deliver their effect through drawdown control rather than return.

A research-grounded roadmap. The strategy literature suggests a concrete, testable programme in which the DER plays its proven role — risk shaping — on top of documented signals.

1. **Orthogonal signal ensemble.** Rather than learning a single signal (which we showed overfits), combine the few signals known to carry out-of-sample information: cross-sectional momentum [19] and time-series momentum [21], which is documented to “perform best during extreme markets.” Each is used with fixed, theory-driven sign, and the ensemble is weighted by *risk parity* [24] rather than by a learned mapping, directly addressing the leverage-aversion and diversification concerns that motivate that construction.
2. **State-dependent θ as principled volatility management.** Momentum’s characteristic failure mode is the *momentum crash* [22], and the established remedy is to scale exposure by ex-ante inverse volatility. This is exactly what a regime-aware, state-dependent risk aversion θ_t implements within the DER: raising θ_t when volatility (or the directionally-uninformative but risk-informative volume signals of Section 7.4) is high de-risks the book before turbulence. Volatility-managed portfolios [23] report higher Sharpe ratios precisely because volatility is more forecastable than returns — giving the DER risk layer a documented mechanism through which to add value.
3. **Honest out-of-sample caveat.** We will test these against the known caution that volatility management can *harm* real-time performance and is most reliable when combined with, rather than replacing, the unmanaged strategy [23]; the evaluation must therefore be strictly walk-forward and report combined as well as stand-alone variants.
4. **Methodological neighbours.** The Sharpe-optimised deep momentum networks of Lim, Zohren and Roberts [25], which learn position sizing within a volatility-scaling framework and add a turnover regulariser, are the closest methodological reference point; replacing their

Sharpe objective with the DER, and their LSTM with our entropic per-step reward, is a natural head-to-head experiment.

This programme keeps the contribution of the present paper intact — the DER as a principled, single-parameter risk layer — while pursuing performance through the only avenue our evidence supports: combining genuine, orthogonal signals under disciplined, volatility-aware risk control.

12 Conclusion

We introduced the Differential Entropic Reward, an online, differentiable, telescoping per-step reward that renders the entropic certainty equivalent — and hence sensitivity to all return cumulants through a single parameter θ — usable by gradient-based RL trading agents. We proved its limiting cases, cumulant expansion, telescoping identity, and robust dual, verified each numerically, and showed empirically on real equities that it controls drawdown decisively and significantly relative to the Differential Sharpe Ratio, while honestly delimiting that it does not manufacture predictive signal. The DER is a principled, interpretable, single-parameter mechanism for the one thing it demonstrably delivers: control of downside risk.

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